Numerical processing of financial data
APM 50442 EP
Part 4: market microstructure

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1 Sequential stock data in one dimension

2 Sequential stock data in higher dimension

3 From sequential data to portfolio selection

4 Market microstructure
- Order book data
- Bid-ask spread
- Liquidity risk
- Market impact

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Types of financial marketplaces (1/2)

1. Walras auctions

The mechanism: An auctioneer gathers supply or demand of each agent for each possible price, then he adjusts the price to get an equilibrium.
Problems:

- possibly no transactions (if buyers and sellers are too greedy)
- or unstable prices (if all sellers are greedy: small volume executed at a very high price).

2. Organized market

Mechanism in two steps:

1. **Quoting**: A **market maker** (a visible market participant), displays volumes and prices he proposes for bid and ask; other agents make auctions (volume and price).
2. **Clearing**: The market maker executes possible bids and asks, consistently with its initial display.

Consequences:

- Displaying market maker's prices avoids auctions being too far from these prices.
- The market maker can build strategies to stabilize the market and maximize the transaction volume.

Types of financial marketplaces (2/2)

3. Electronic exchange

- Continuous double auction: transaction is done as soon as a buyer and a seller agree on a price.
- **Limit order book** is public (for lit markets, not dark pools) and dynamic: it contains limit orders, i.e. orders not yet executed.
- No more market maker but **liquidity providers** (patient traders) and **liquidity takers** (impatient traders).
- Predefined grid for prices (price unit: tick size) and volumes (volume unit: lot size).
- Main types of orders issued:
  - if immediately executed: marker order.
  - if not: limit order.
  - cancellation orders.
  - hidden orders: so that large orders will not lead to adverse reaction of competitors.
- buy order = bid $<$ ask = sell order.
- bid-ask spread = ask - bid > 0.
- mid price $= \frac{1}{2}$ (bid + ask).

Order book (1/5)

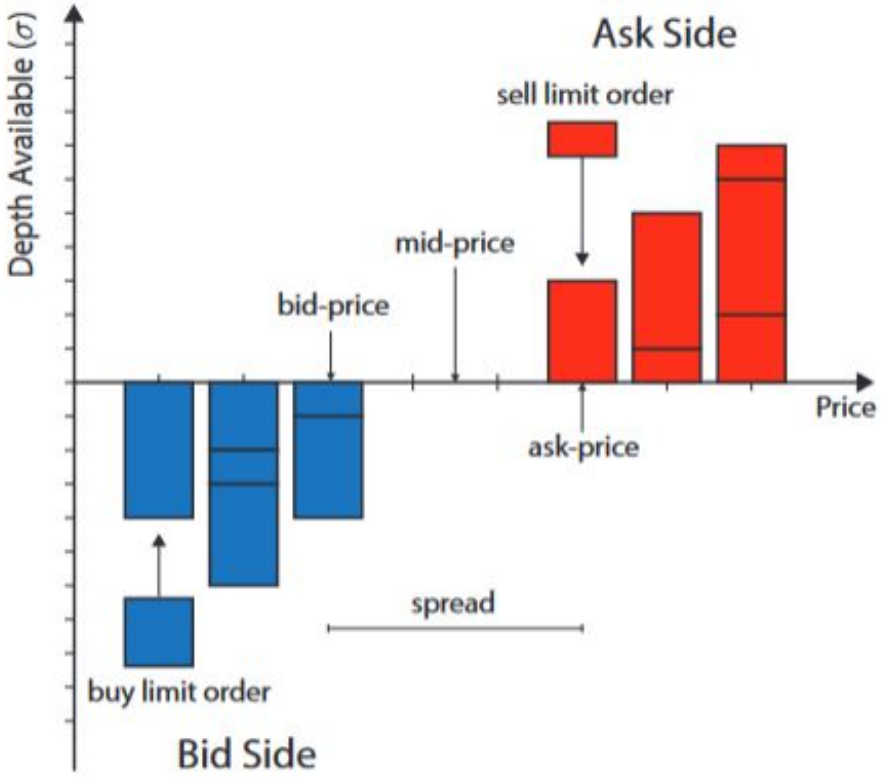

Figure 1: Schematic of an LOB. The horizontal lines within the blocks at each price level denote the different active orders at each price.

(Source: Bonart, Gould)

Order book (2/5)

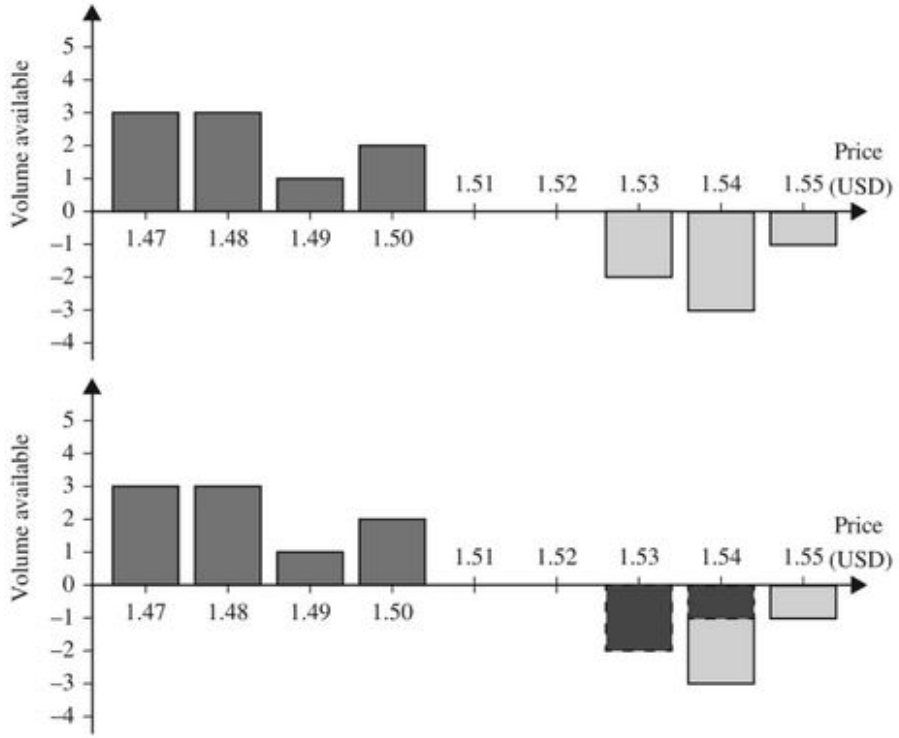

Figure 3.3. An illustration of how the arrival of an order $(+1, \$1.55, 3, t_x)$, as described by the third line of Table 3.1, would impact the LOB shown in Figure 3.2. The dashed boxes denote the limit orders that match to the incoming market order, and that are therefore removed from the LOB. The ask-price immediately after the market order arrival is $\overline{a}(t_x) = \$1.54$.

Table 3.1. *Example to illustrate how specified order arrivals would affect prices in the LOB in the top panel of Figure 3.3. The tick size is $\vartheta = 0.01$ and the lot size is $v_0 = 1$ (see Section 3.1.5).*

Arriving order x	Values before arrival (USD)				Values after arrival (USD)			
$(\varepsilon_x, p_x, v_x, t_x)$	$b(t_x)$	$a(t_x)$	$m(t_x)$	$s(t_x)$	$\overline{b}(t_x)$	$\overline{a}(t_x)$	$\overline{m}(t_x)$	$\overline{s}(t_x)$
$(+1, \$1.48, 3, t_x)$	1.50	1.53	1.515	0.03	1.50	1.53	1.515	0.03
$(+1, \$1.51, 3, t_x)$	1.50	1.53	1.515	0.03	1.51	1.53	1.52	0.02
$(+1, \$1.55, 3, t_x)$	1.50	1.53	1.515	0.03	1.50	1.54	1.52	0.04
$(+1, \$1.55, 5, t_x)$	1.50	1.53	1.515	0.03	1.50	1.55	1.525	0.05
$(-1, \$1.54, 4, t_x)$	1.50	1.53	1.515	0.03	1.50	1.53	1.515	0.03
$(-1, \$1.52, 4, t_x)$	1.50	1.53	1.515	0.03	1.50	1.52	1.51	0.02
$(-1, \$1.47, 4, t_x)$	1.50	1.53	1.515	0.03	1.48	1.53	1.505	0.05
$(-1, \$1.50, 4, t_x)$	1.50	1.53	1.515	0.03	1.49	1.50	1.495	0.01

(Source: Bouchaud, Bonart, Donier, Gould)

A limit or a market order x is characterized by $(\varepsilon_x, p_x, v_x, t_x)$, with $\varepsilon_x \in \{-1, +1\}$ the direction of the trade (buy $= +1$), p_x the price, v_x the volume, t_x the time.

Order book (3/5)

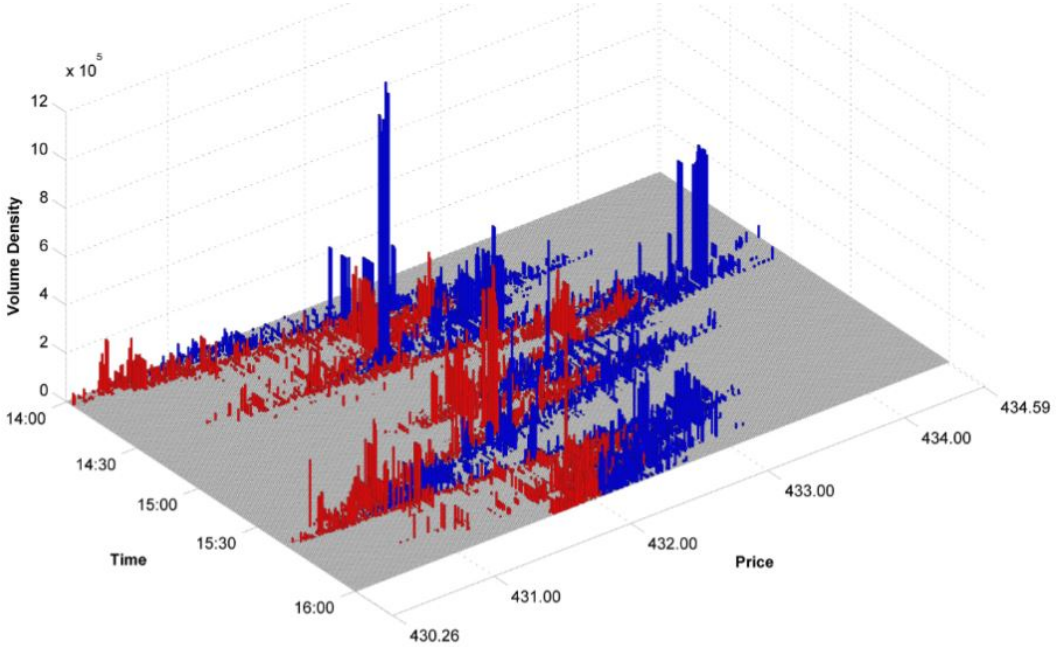

Figure 1.6: The order book of AAPL from the 4th of April 2013 between 14:00 and 16:00, 20 price levels from the spread, sampled every 300 events.

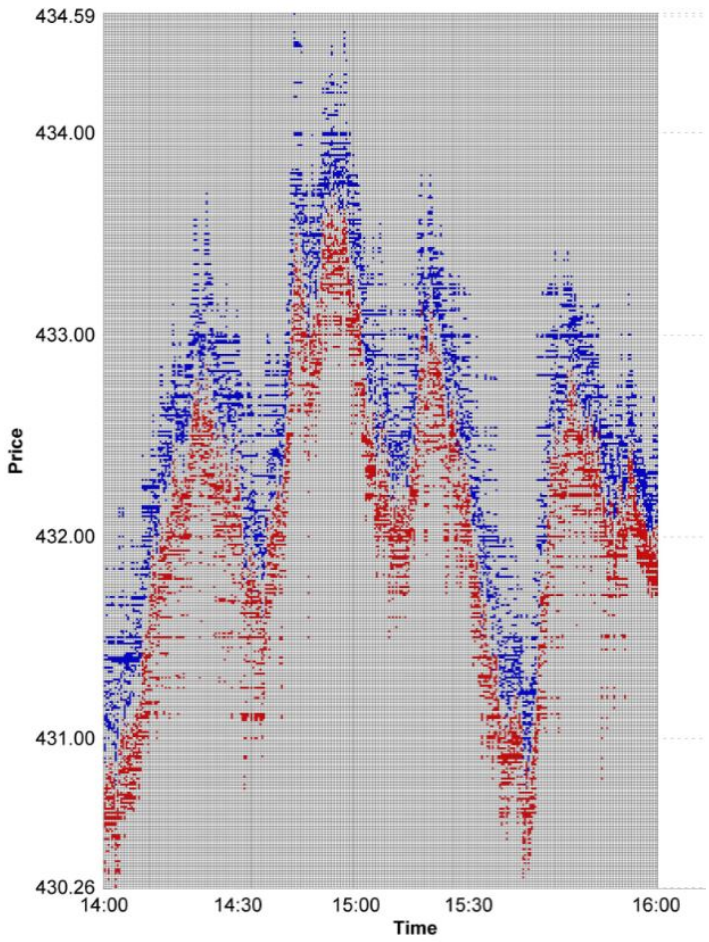

Figure 1.7: The plotted price evolution of the AAPL order book in Figure 1.6.

(Source: Paulsen)

Order book (4/5)

Some useful statistics to describe the order book, for bid prices $b_1 > b_2 > \ldots$ and ask prices $a_1 < a_2 < \ldots$ of volumes $v(b_i)$ and $v(a_i)$:

- The cumulative buying and selling volume:

$$\mathcal{V}^{\text{bid}}(p) = \sum_i v(b_i)\mathbb{1}_{b_i \geq p} \qquad \text{and} \qquad \mathcal{V}^{\text{ask}}(p) = \sum_i v(a_i)\mathbb{1}_{a_i \leq p}.$$

- The buying and selling market depth:

$$\sum_i v(b_i) = \lim_{p \to -\infty} \mathcal{V}^{\text{bid}}(p) \qquad \text{and} \qquad \sum_i v(a_i) = \lim_{p \to +\infty} \mathcal{V}^{\text{ask}}(p).$$

- The volume imbalance:

$$\frac{\sum_i v(b_i) - \sum_i v(a_i)}{\sum_i v(b_i) + \sum_i v(a_i)}.$$

- The order book bid-ask spread: best ask - best bid $= a_1 - b_1$.

$\rightsquigarrow$ Python exercise, Question 1

Order book (5/5)

https://www.binance.com/en/orderbook/BTC_USDT

BINANCE | Buy Crypto | Markets | Trade | Futures | Earn | Square | More | Log In

Exchange > Order Book

Order Book — BTC/USDT

Depth: 15 | Group: 2 decimals

Buy Order

Side	Price (USDT)	Amount (BTC)	Total (USDT)	Sum (USDT)
Buy 1	107,029.20	0.15360	16,439.6851200	16,439.6851200
Buy 2	107,029.10	0.00015	16.0543650	16,455.7394850
Buy 3	107,028.84	0.00456	488.0515104	16,943.7909954
Buy 4	107,028.28	0.00025	26.7570700	16,970.5480654
Buy 5	107,028.27	0.03435	3,676.4210745	20,646.9691399
Buy 6	107,028.13	0.00015	16.0542195	20,663.0233594
Buy 7	107,028.00	0.00710	759.8988000	21,422.9221594
Buy 8	107,027.44	0.00049	52.4434456	21,475.3656050
Buy 9	107,027.06	0.00015	16.0540590	21,491.4196640
Buy 10	107,026.71	0.00010	10.7026710	21,502.1223350

Sell Order

Side	Price (USDT)	Amount (BTC)	Total (USDT)	Sum (USDT)
Sell 1	107,029.21	1.79128	191,719.2832888	191,719.2832888
Sell 2	107,029.44	0.24988	26,744.5164672	218,463.7997560
Sell 3	107,029.45	0.00010	10.7029450	218,474.5027010
Sell 4	107,029.46	0.00010	10.7029460	218,485.2056470
Sell 5	107,029.47	0.00010	10.7029470	218,495.9085940
Sell 6	107,029.48	0.00010	10.7029480	218,506.6115420
Sell 7	107,029.49	0.16575	17,740.1379675	236,246.7495095
Sell 8	107,029.80	0.00685	733.1541300	236,979.9036395
Sell 9	107,029.81	0.00010	10.7029810	236,990.6066205
Sell 10	107,029.82	0.00010	10.7029820	237,001.3096025

$\rightsquigarrow$ Can you identify the above statistics in this LOB?

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Measuring bid-ask spreads

- Displayed price P_t is the price of the last transaction (bid or ask). Bid/ask alternation creates price movements even if the mid price $P_t^\star$ is constant:

$$P_t = P_t^\star(1 + S(X_t - 0.5)),$$

where $X_t \sim \mathcal{B}(1/2)$ is the trade direction.

- At a microscopic scale, additive noise is preferred to multiplicative one or we can also consider that p_t is now the logarithmic price:

$$p_t = p_t^\star + \eta_t,$$

where $\eta_t = S(X_t - 0.5)$ describes the microstructure noise.

- For a limit order book, S is observed.
- Not the case for illiquid markets (corporate bonds), OTC markets, auction markets, dark pools.
- For a liquid and lit market, how can one calculate a low-resolution aggregation of the series of bid-ask spreads?

Standard estimators

Assumptions

- Increments $p_t^\star - p_s^\star$ are Gaussian with variance parameter $\sigma^2|t-s| > 0$,
- Non-overlapping increments of $p_t^\star$ are independent of each other.
- η_s and η_t are independent of each other as soon as $s \neq t$.
- $p_t^\star$ and η_t are independent of each other.

Estimators:

- Roll estimator, using empirical covariances $\widehat{C}_n(p_t, \tau, k) = \frac{1}{n-2-k} \sum_{i=0}^{n-3-k} \left(p_{(i+1)\tau} - p_{i\tau}\right) \left(p_{(i+2+k)\tau} - p_{(i+1+k)\tau}\right)$:

$$\mathrm{Roll}_n^2 = -4\widehat{C}_n(p_t, \tau, 0).$$

- An estimator based on empirical variances $\widehat{V}_n(p_t, \tau) = \widehat{C}_n(p_t, \tau, -1)$:

$$\widehat{S}_n^2 = 2\left(2\widehat{V}_n(p_t, \tau) - \widehat{V}_n(p_t, 2\tau)\right).$$

$\rightsquigarrow$ Why are these estimators consistent estimators of S^2?

Unbiased estimator for autocorrelated trades

Alternative assumptions

- Increments $p_t^\star - p_s^\star$ are Gaussian with variance parameter $\sigma^2|t-s| > 0$,
- Non-overlapping increments of $p_t^\star$ are independent of each other.
- η_s and η_t are correlated, with a correlation $\exp\left(-\frac{|t-s|}{\lambda}\right)$.
- $p_t^\star$ and η_t are independent of each other.

Bias in traditional estimators

- For the two estimators: the bias is $-S^2(2e^{-\tau/\lambda} - e^{-2\tau/\lambda})$. $\rightsquigarrow$ Prove it.
- Relative bias for S^2 of -44% when λ is such that half life of 30 seconds, -75% for a half life of 1 minute, for $\tau = 1$ min.

A new estimator: $\rightsquigarrow$ Is it consistent?

$$\widetilde{S}_n^2 = \frac{2\left(2\widehat{V}_n(p_t,\tau) - \widehat{V}_n(p_t,2\tau)\right)^2}{\left(2\sqrt{\left|2\widehat{V}_n(p_t,\tau) - \widehat{V}_n(p_t,2\tau)\right|} - \sqrt{\left|2\widehat{V}_n(p_t,2\tau) - \widehat{V}_n(p_t,4\tau)\right|}\right)^2}.$$

Unbiased estimator for autocorrelated trades

$\mathbb{S}$	Bias	Standard dev.	Quadratic risk	test $\mathbb{S} = S$ (p-value)
Roll	$\mathbf{-6.6 \times 10^{-7}}$	2.9×10^{-4}	8.2×10^{-8}	**Accepted** (1.0×10^{0})
$\widehat{S}$	3.9×10^{-6}	$\mathbf{1.8 \times 10^{-4}}$	$\mathbf{3.1 \times 10^{-8}}$	**Accepted** (9.8×10^{-1})
$\widetilde{S}$	3.1×10^{-4}	9.2×10^{-4}	9.5×10^{-7}	**Accepted** (7.3×10^{-1})

$\mathbb{S}$	Bias	Standard dev.	Quadratic risk	test $\mathbb{S} = S$ (p-value)
Roll	-1.6×10^{-3}	3.1×10^{-4}	2.8×10^{-6}	Rejected (2.0×10^{-7})
$\widehat{S}$	-1.3×10^{-3}	$\mathbf{2.2 \times 10^{-4}}$	1.6×10^{-6}	Rejected (2.3×10^{-8})
$\widetilde{S}$	$\mathbf{3.3 \times 10^{-4}}$	1.1×10^{-3}	$\mathbf{1.4 \times 10^{-6}}$	**Accepted** (7.7×10^{-1})

Table: For $S = 0.5\%$ and 10,000 simulated trajectories of one trading day sampled every minute (High and Low based on a one-second sampling) with **uncorrelated (top) or correlated (bottom) trades**, bias, standard deviation, and quadratic risk of the estimators. The last column is the output of a statistical test whose null hypothesis is that the estimator $\mathbb{S} \in \{\text{Roll}, \widehat{S}, \widetilde{S}\}$ is equal to S, with a confidence of 95%, according to the simulated distribution of the estimator (corresponding p-value in parenthesis).

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Liquidity: confronting many viewpoints

The bid-ask spread (difference of price between seller and buyer) can be split into two from the **market maker**'s perspective:

- **Liquidity cost**: the real spread that compensates the market maker for his role as guarantor of market liquidity and which corresponds to the technology and inventory costs incurred by the market maker.
- **Market maker risk**: the information asymmetry that compensates market makers for the losses they might incur when they make a bet without knowing its ins and outs: since they cannot distinguish between informed and uninformed bets, they can be the "opposite part" of an informed order because of their role as guarantor of market liquidity.

Liquidity: confronting many viewpoints

According to Amihud and Mendelson (2006), there would be three components in liquidity costs:

- **Direct trading costs**: deterministic transaction costs that include brokerage commissions, transaction taxes and exchange fees.
- **The impact of orders on the market price**: all the more important as the order is substantial, because a large order will involve several counterparts and therefore go far in the order book.
- **The cost induced by a delay in execution**: to limit the impact of a large order, traders can do block orders, which means that some blocks are exchanged with a certain delay, which exposes more to the possibility of market movements.

Liquidity: confronting many viewpoints

According to Harris (1990), the notion of a liquid market is defined as a market on which a large quantity can be bought and sold at any time, quickly and at low cost. It is measured by four pillars:

- **Width**: cost incurred by a simultaneous purchase/sale, which corresponds to the sum of direct costs and impact in Amihud and Mendelson.
- **Depth**: the quantity of securities that can be traded without impact (only cost: bid/ask spread) because absorbed without difficulty by the market maker (see following graph).
- **Immediacy**, which determines the speed with which positions can be traded and which corresponds to the delay between the placing of the order and its settlement.
- **Resilience**, which designates the market's capacity to absorb random shocks, such as non-informative orders.

Liquidity: confronting many viewpoints

Liquidity regimes

According to Stange and Kaserer (2009), liquidity costs can be defined continuously and thus form four degrees of liquidity:

- **Cost-free trading** (e.g. cash).
- **Continuous trading**: orders executed at a given cost.
- **Interrupted trading**: orders executed punctually.
- **No trading**: the market is completely illiquid, prices are not available and are recovered by adapted techniques

- Liquidity risk cannot be diversified: no hedging possible, no liquidity derivatives, etc.
- What is the price of an asset when it is not traded? Can we use a comparison with a liquid universe? (example of yield curves for corporate bonds)
- Liquidity premium: positive link between yield and illiquidity (Amihud and Mendelson) but concave (yields increase less for very illiquid assets).
- For corporate bonds, illiquidity increases when the credit rating deteriorates (risk premium of 0.6% in IG [investment-grade bonds], 1.5% in HY [high-yield bonds]).

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Market impact and slippage

Definitions

- The **market impact** is the effect of a trade on the price of an asset. It can be decomposed in permanent and transient impact.
- The **slippage** is a difference between the execution price of a trade and the price displayed just before the transaction. It can be caused by:
  1. High volatility or execution delay: the natural dynamic of prices make them change between order and execution.
  2. Low liquidity or big order: the best price does not offer a large enough volume and additional volumes are to be found in the limit order book.

$\rightsquigarrow$ Python exercise, Question 2

Optimal execution: Almgren and Chriss (1/5)

Almgren and Chriss: optimal liquidation framework (metaorder: take time to limit impact *vs* hurry to limit market risk).

- Quantity X to be liquidated in tranches $n_1, n_2, \ldots$, with $x_k = X - \sum_{j=1}^{k} n_j$.
- Discrete-time trading, time step τ.
- Price dynamics according to arithmetic random walk: $S_k = S_{k-1} + \sigma\sqrt{\tau}\varepsilon_k - \tau g(n_k/\tau)$, where g is the permanent impact function, depending on the liquidation speed n_k/τ between times t_{k-1} and t_k.
- Let's assume that observed price is different from S (underlying dynamics) due to transitory impact (trader drains order book between t_{k-1} and t_k, but it instantly replenishes in t_k): $\bar{S}_k = S_k - h(n_k/\tau)$.
- Price obtained for liquidation (initial frictionless price + illiquidity losses):

$$\sum_{k=1}^{N} n_k \bar{S}_k = XS_0 + \sum_{k=1}^{N} \left[\left(\sigma\sqrt{\tau}\varepsilon_k - \tau g\left(\frac{n_k}{\tau}\right) \right) x_k - n_k h\left(\frac{n_k}{\tau}\right) \right].$$

Optimal execution: Almgren and Chriss (2/5)

- Problem: minimize expected value and variance of the loss with ε_k the only random variables: $E(x) + \lambda V(x)$, with:

$$\left\{ \begin{array}{rcl} E(x) & = & \sum_{k=1}^{N} \left[\tau x_k g\left(\frac{n_k}{\tau}\right) + n_k h\left(\frac{n_k}{\tau}\right) \right] \\ V(x) & = & \sigma^2 \sum_{k=1}^{N} \tau x_k^2. \end{array} \right.$$

- Almgren and Chriss hypothesis: linear impact:
  - $g(v) = \gamma v$, which leads to a total cost

$$\begin{array}{rcl} \sum_{k=1}^{N} \tau x_k g\left(\frac{n_k}{\tau}\right) & = & \gamma \sum_{k=1}^{N} x_k (x_{k-1} - x_k) \\ & = & \frac{1}{2}\gamma \sum_{k=1}^{N} x_{k-1}^2 - x_k^2 - (x_k - x_{k-1})^2) \\ & = & \frac{1}{2}\gamma X^2 - \frac{1}{2}\gamma \sum_{k=1}^{N} n_k^2. \end{array}$$

  - $h(n_k/\tau) = \xi sgn(n_k) + \eta n_k/\tau$, with ξ being half the bid-ask spread. This model for h is often called the quadratic cost model because the associated cost is $n_k h(n_k/\tau) = \xi |n_k| + \eta n_k^2/\tau$.

Optimal execution: Almgren and Chriss (3/5)

- Minimum impact by taking constant speed: $n_k = X/N$, but variance can be high.
- Minimum variance by liquidating everything at the first instant: $n_1 = X$ and $n_2 = ... = n_N = 0$. Then, $V = 0$ and $E = \xi X + \eta X^2/\tau$.
- Between the two: an efficient frontier for several levels of risk aversion corresponding to the Lagrangian λ:

$$x_k = \frac{sinh(K(T-k))}{sinh(KT)} X,$$

where $K \overset{\tau \to 0}{\sim} \sqrt{\frac{\lambda \sigma^2}{\eta}} + \mathcal{O}(\tau)$.
$\rightsquigarrow$ How do we get this equation?

Optimal execution: Almgren and Chriss (4/5)

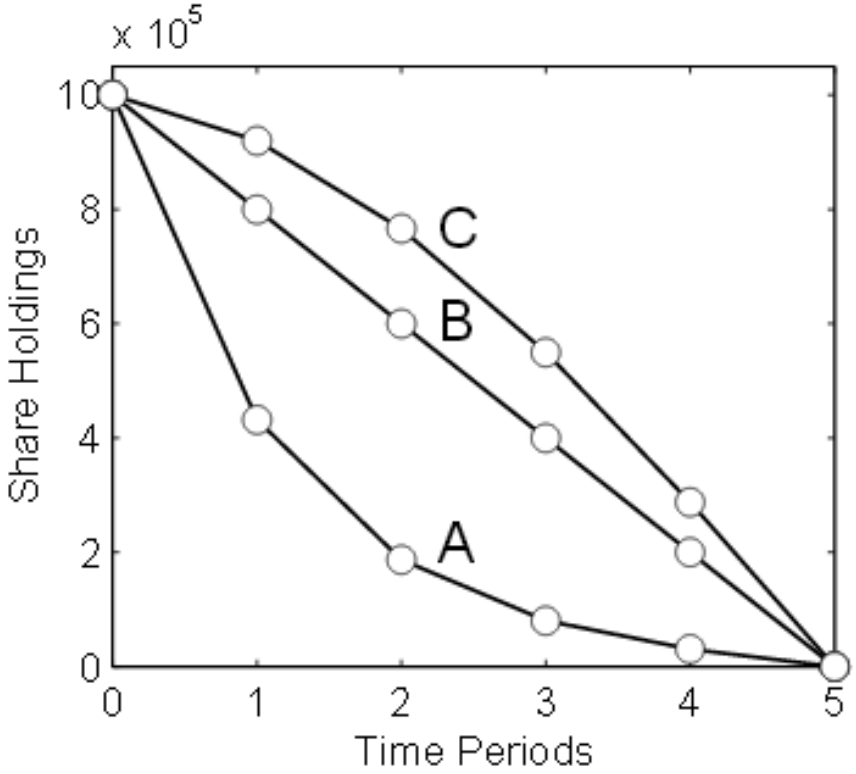

Figure 2: *Optimal trajectories.* The trajectories corresponding to the points shown in Figure 1. (A) $\lambda = 2 \times 10^{-6}$, (B) $\lambda = 0$, (C) $\lambda = -2 \times 10^{-7}$.

Figure: x for various risk aversions. Source : Almgren et Chriss.

Optimal execution: Almgren and Chriss (5/5)

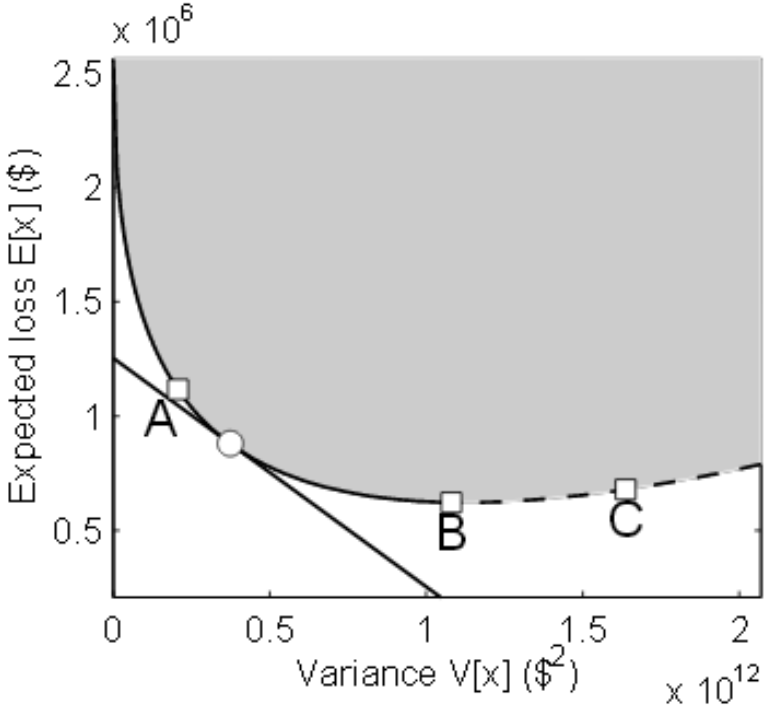

Figure 1: *The efficient frontier.* The parameters are as in Table 1. The shaded region is the set of variances and expectations attainable by some time-dependent strategy. The solid curve is the efficient frontier; the dashed curve is strategies that have higher variance for the same expected costs. Point B is the "naïve" strategy, minimizing expected cost without regard to variance. The straight line illustrates selection of a specific optimal strategy for $\lambda = 10^{-6}$. Points A,B,C are strategies illustrated in Figure 2.

Figure: Source : Almgren et Chriss.

Market impact: econophysics approach

Backward definition of the price: the price is the sum of past impacts (Bouchaud):

$$p_t = p_{-\infty} + \sum_{s=-\infty}^{t-1} G(t-s)\varepsilon_s S_s V_s^r,$$

where:

- S is the bid-ask spread;
- ε is -1 or 1 depending on whether the transaction is a buy (price = ask) or a sell (price = bid) on the market;
- V is the volume of the transaction, with r close to zero to have a concave function of the volume;
- G is a function worth 0 on $\mathbb{R}^-$, which can be interpreted as the impact of a single order. A statistical study on long-term correlation can allow this function to be fixed, in the form, for example, of a decreasing power function.